10 APRIL 2025

LIVERPOOL HOPE UNIVERSITY

1844

BIG DATA AND CLOUD COMPUTING COURSEWORK 1

DATA ANONYMISATION IN HEALTHCARE

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1. Introduction to Data Anonymisation in Healthcare:

Data anonymisation is an important process in healthcare because it helps protect patient information while still allowing useful research and analysis. The main goal of anonymisation is to remove or change personal information so that no one can identify the patient. This is done by following privacy rules, such as only keeping the information that is necessary and making sure sensitive details are hidden. Common methods of anonymisation include masking, pseudonymisation, and other techniques that keep data useful for research but secure from unauthorized access.

1.1 Data Anonymisation Principles and Techniques:

One key idea in data anonymisation is data minimisation. This means that only the most necessary information should be kept in its original form. For example, personal details like names, addresses, and phone numbers are often removed or replaced. Some information, like a person's birth date or location, can still indirectly reveal their identity, so it is changed to make it harder to identify the person. Techniques like masking and tokenisation help ensure that even if data is accessed, it cannot be linked back to the person. (Colombi, 2023)

Another method is pseudonymisation, where real identifiers are replaced with fake or temporary ones. This allows the data to still be useful for analysis while keeping sensitive information separate and protected. Many healthcare systems use a combination of these methods to balance keeping data useful for research while also protecting patient privacy. These techniques need to be updated regularly to keep up with new data privacy challenges. (Aamot et al., 2013)

1.2 Legal, Ethical, and Regulatory Considerations:

In the European Union, the General Data Protection Regulation (GDPR) is the key legislation. It defines personal data as any information that can identify a person, directly or indirectly. GDPR allows for the processing of personal health data under strict conditions but exempts anonymised data from its scope—provided that the anonymisation is irreversible and the individual can no longer be identified by any means. (Data Protection Commission, 2019)

In the United States, the Health Insurance Portability and Accountability Act (HIPAA) includes a Privacy Rule that outlines two methods for de-identifying data: Expert Determination and Safe Harbor. The Safe Harbor method requires the removal of 18 types of identifiers (e.g.,

names, full-face photos, geographic data smaller than a state level, etc.) for the data to be considered de-identified. (Centers for Disease Control and Prevention, 2024) (Alder, 2024)

From an ethical perspective, healthcare providers must protect patient confidentiality. This responsibility goes beyond simply following the law; it is also about building trust with patients. Patients are more likely to share important information if they feel confident that their data will be protected. It is essential to be open about how data is anonymised and how it is used to ensure that patients understand and trust the process. (Olorunfemi et al., 2024)

1.3 Cloud Computing for Anonymising Large Patient Datasets:

As healthcare data grows—especially with electronic health records and diagnostic images—traditional methods of handling and anonymising this information can be slow and inefficient. Cloud computing offers a better solution. It can handle large amounts of data quickly and efficiently, and it is flexible, meaning it can adjust to the needs of healthcare institutions. (Sudha et al., 2024) Cloud platforms come with strong security features and meet international data protection standards, which makes them great for running anonymisation processes. (Rafalski, 2025)

Using cloud services means data can be processed in real time and monitored continuously, which helps improve performance and reliability. These platforms can also adjust resources as needed to handle large amounts of data, which is especially useful for healthcare organizations that need to update records often. Additionally, cloud environments can work with powerful anonymisation tools, creating a safe and effective way to manage large patient datasets while still following privacy laws and keeping the data secure. (Sudha et al., 2024) (Rafalski, 2025)

2 Data Anonymisation Techniques and Algorithms:

Today, we use different methods and computer tools to change sensitive data so it is not easy to identify. These methods are different in how hard they are to use, how much privacy they give, and how useful the data stays after anonymisation.

2.1 Substitution Algorithms:

Substitution means changing real names or details with fake but believable ones. For example, "John Smith" becomes "David Miller", but we keep the same medical info like test results or diagnosis. This is helpful for training computer models because the data still looks real. In hospitals, it helps to train systems without using real patient names. One problem is that if the

rules for substitution are easy to guess, someone could still find patterns in the data. (IBM, 2018)

2.2 Randomized Algorithms:

Randomization means mixing up or changing values using math. For example, a birth date like "1990-05-15" could become "1988-11-22". A strong example is differential privacy, which adds noise to answers so that people cannot guess individual data, but total numbers still look correct. Hospitals use this when they give infection data to health organizations. The hard part is adding just the right amount of noise — not too much or too little — so the data is still useful. (Bose, Roy, & Naskar, 2020)

2.3 Numerical Variation:

This method changes numbers a little bit, like changing lab results by ±10%. For example, 14.2 g/dL could become 13.7 g/dL. Studies show that if the change is more than 15%, the data might not be useful anymore. Drug companies use this method to study side effects from different hospitals, so they can still see trends but not exact values.

2.4 k-Anonymity:

k-Anonymity means that each person in the data looks the same as at least (k-1) others. If k=5, then every group of details (like age, location, and diagnosis) must appear in at least five records. Hospitals use this for COVID-19 data by grouping ages (like 20–29, 30–39) and hiding rare postcodes. This helps stop simple ways to find people, but it doesn't work well when there are many details that make people unique. (Ok, Bright and Joshua, 2023)

2.5 l-Diversity:

This is an improvement of k-anonymity. It says that every group must have at least l different values for private data. For example, if l=3 in a cancer dataset, each group must have at least three different tumour stages. This stops people from guessing someone's medical problem just by knowing their age and postcode. It's useful in clinical trials when sharing data safely. (Ok, Bright and Joshua, 2023)

2.6 t-Closeness:

t-Closeness makes l-diversity even stronger. It says that the private data in each group should look almost the same as the whole dataset, not too different (within a small limit t). This helps when the data has unusual patterns. For example, national health groups use t-closeness to share

disease numbers from different areas, making sure no area shows very different results than the rest of the country. (Ok, Bright and Joshua, 2023)

2.7 Pseudonymization:

Pseudonymization changes names or ID numbers into secret codes using special math (like cryptographic hashes). For example, a number like "9876543210" becomes "X9b4#qW2" using SHA-256 and a secret key. Only people with permission can turn it back to the real number. Hospitals in Europe use this method to work together on cancer research. It helps follow rules like GDPR. But it's not full anonymisation, because there's still a way to find out who the person is, so strong security is important. (Aamot et al., 2013)

2.8 Tokenization:

Tokenization means changing private data into a special code called a token. This token does not show anything about the original data and cannot be changed back without a secure system called a token vault. This is different from pseudonymization, because tokens do not follow a pattern, so no one can guess the real data without the vault. For example, a Social Security Number like "123-45-6789" becomes "TKN-7X2B9". This method is very useful in telehealth apps, where payment data needs to be safe but still checked later for records or reports. (Colombi, 2023)

3. Application Solution:

3.1 Assumptions and Design Considerations

The application is designed to anonymize sensitive patient information from medical notes. Several assumptions and design choices shape its development:

- **Sensitive Data Types:**
  The application focuses on anonymizing four key data types:
  - **Names:** With or without titles like Mr., Dr.
  - **Ages:** Two-digit numbers with keywords like "years old" and "aged"
  - **Addresses:** Simplified formats with street numbers, names, and phrases like "residing at"
  - **Dates of Birth:** In all different formats like dd/mm/yyyy, yyyy-mm-dd, etc.

- **Mapping and Identification:**
  Each sensitive data item is replaced with a unique token in the format <patientNumber>.<attributeCounter>. These are stored in a LinkedHashMap to maintain the order of appearance, which helps generate a clear and accurate mapping document.

3.2 Regex Patterns and Matchers:

The application employs several regex patterns designed to capture different forms of sensitive information:

3.2.1 Name Pattern:

```
\\b(?:(?:Mr\\.?|Mrs\\.?|Ms\\.?|mr\\.?|mrs\\.?|ms\\.?|Dr\\.?|dr\\.?|Prof\\.?|[Mm]\\.?|[Dd]\\.?)\\s+[A-Z][a-zA-Z'\\-]+(?:\\s+[A-Z][a-zA-Z'\\-]+)*|[A-Z][a-zA-Z'\\-]+(?:\\s+[A-Z][a-zA-Z'\\-]+)+)\\b
```

- **Rationale:**
  - **Dual Alternatives:** The pattern has two branches:
    - **With Titles:** This regex looks for common name titles like "Mr", "Dr", or "Prof", even if they are written with small mistakes like missing dots or lowercase letters(like, “mr.”, “dr.”). It then matches two or more capitalized words after the title to find full names.
    - **Full Names without Titles:** This branch expects at least two capitalized words, reducing false positives from common words.
  - **Character Flexibility:** Inclusion of apostrophes and hyphens gives flexibility for last names like “DrinkingWater” or double-barrelled surnames.
    - **Post-Match Processing:** After a match, the code further splits the name (after removing any titles) to capture individual parts, so that repeated occurrences of the same name parts anywhere in the text are also anonymized.

3.2.2 Age Pattern:

```
\\b(\\d{2})\\s*?yearold\\b|\\bage:\\s*(\\d{2})\\b|\\b(\\d{2})\\s*years?\\s*old\\b|\\baged\\s*(\\d{2})\\b
```

- **Rationale:**
    - **Multiple Formats:** The pattern accounts for various common expressions for age (e.g., “45 year-old”, “age: 45”, “45 years old”, “aged 45”).

3.2.3 Address Pattern:

\\d+\\s+[A-Za-z\\s]+(?:,\\s*[A-Za-z\\s]+)?(?:,\\s*[A-Za-z]{2})\\b(?:,\\s*\\d{5})?|\\b(residing|living)\\s+(at|in)\\s+[A-Z][a-zA-Z]*(?:\\s+[A-Z][a-zA-Z]*)*,?\\s+[A-Z]{2}\\b|\\bresident of\\s+[A-Z][a-zA-Z]*(?:\\s+[A-Z][a-zA-Z]*)*,?\\s+[A-Z]{2}\\b

- **Rationale:**
    - **Variations in Address Format:** The regex covers numeric street addresses (with optional state abbreviations and zip codes) while also handling phrases like “residing at” or “resident of”.
    - **State Abbreviations:** It even expects and matches state abbreviations (two capital letters) to help identify location markers.

3.2.4 DOB Pattern:

\\b(?:(?:0?[1-9]|[12][0-9]|3[01])[-/.](?:0?[1-9]|1[0-2])[-/.](?:\\d{2}|\\d{4})|(?:0?[1-9]|1[0-2])[-/.](?:0?[1-9]|[12][0-9]|3[01])[-/.](?:\\d{2}|\\d{4})|(?:\\d{4})[-/.](?:0?[1-9]|1[0-2])[-/.](?:0?[1-9]|[12][0-9]|3[01]))\\b

- **Rationale:**
    - **Multiple Date Formats:** It handles several common date formats such as dd-mm-yyyy, mm/dd/yyyy, or even yyyy.mm.dd.
    - **Validation Scope:** While it checks for plausible day and month ranges, it isn’t a full validation for leap years or month-specific day limits, which is acceptable for anonymization purposes where exact validation is less critical than identification.

3.3 Java Implementation Overview

The complete Java implementation is organized into a few key components:

3.3.1 Main Application Flow:

- **User Input and File Handling:**
  - The main method prompts the user for an input file path.
  - Output file paths for the anonymized notes and mapping document are derived from the parent directory of the input file.
- **Core Anonymization Process:**
  - The anonymizeMedicalNotes method reads the complete file content and splits it into separate patient records.
  - Each record is processed individually using the anonymizePatientRecord method.

3.3.2 Anonymizing a Patient Record:

- **Regex-based Matching:**
  - Within anonymizePatientRecord, the sensitive fields are identified using the regex patterns discussed earlier.
  - Sensitive data such as names, ages, addresses, and DOBs are stored along with their locations (start and end indices) via the MatchData class.
- **Overlap Resolution and Replacement:**
  - All collected matches are sorted to avoid overlapping issues.
  - A unique identification scheme is applied so that each distinct piece of sensitive data is replaced with a unique token (e.g., "1.1", "1.2", etc.).
  - A mapping document is created by keeping track of each token and its original value.

3.3.3 File I/O and Mapping Document:

- **File Reading and Writing:**

- Utility methods readFile and writeFile perform the actual reading from and writing to disk.
- The writeMappingDocument method outputs a mapping text file that shows which token corresponds to which piece of sensitive data.

3.4 Testing Method and Sample Results:

3.4.1 Testing Methods:

I. Unit Testing of Individual Components:

- **Regex Pattern Validation:**
  Each regex pattern can be independently tested using JUnit tests. For example, you can create tests where you supply different strings and verify the captured groups:
  - A test for names that includes variations with titles (e.g., "Mr. John Doe", "Dr. Samantha Lee") including verifying the special-case exclusion.
  - A test for the age pattern where strings like "45 year-old" or "age: 45" yield correct matches.
  - A test for addresses and DOBs ensuring that the various formats are recognized correctly.

II. Integration Testing:

- **Simulated Patient Records:**
  - Create a sample patient notes file containing multiple records that mix names, ages, and addresses in different formats. For example:
  - Ms. Sarah Evans, 55-year-old, resident of 321Elm Street, Chicago, IL
  - Mr. Joe DrinkingWater, 55-year-old, 55 Penny Lane, Liverpool, UK
- **Execution and Output Verification:**
  Running the application on the sample file should generate:
  - An anonymized file where sensitive data are replaced with identifiers like 1.1, 1.2, etc.
  - A mapping document that details which token maps to each piece of sensitive information.

III. Edge Case Testing:

- **Overlapping Matches:**

  Ensure that if the same name (or its part) appears multiple times in a record, it gets replaced by the same identifier.

- **Missing or Malformed Data:**

  Test what happens if the record does not include one or more sensitive fields.

3.4.2 Sample Results:

```
/home/24014410/PatientNotes.txt
$ java PatientDataAnonymisation
Enter the path to the patient notes file: /home/24014410/PatientNotes.txt
Anonymization completed successfully.
Anonymized notes saved at: /home/24014410/anonymized_notes.txt
Mapping document saved at: /home/24014410/mapping_document.txt
$ ls
'PatientDataAnonymisation$AnonymizationResult.class'   PatientDataAnonymisation.class   PatientNotes.txt       mapping_document.txt
'PatientDataAnonymisation$MatchData.class'             PatientDataAnonymisation.java    anonymized_notes.txt
$
```

- **Anonymized Output:**

```
1.1, a 1.2 female resident of 1.3, presented with mild chest discomfort and occasional shortness of breath during exertion. She has been adhering to a daily dose of am
lodipine (5mg) for hypertension management since 2012, with no significant cardiac history. 1.4, a former smoker of 15 years, now limits herself to occasional social s
moking and moderate alcohol consumption. Family history indicates a prevalence of hypertension and cardiovascular diseases, particularly on the maternal side. Recommen
dations for further evaluation, including ECG and stress test, were made, along with lifestyle modifications to enhance cardiovascular health.
During the consultation, 2.1, a 2.2 resident of 2.3, presented with mild chest discomfort and occasional shortness of breath during physical activity. He has been mana
ging hypertension with a daily dose of amlodipine (5mg) since 2015, with no reported cardiac anomalies or major surgeries. 2.4's 30-year smoking history and moderate a
lcohol consumption pose significant risks to his cardiovascular health. Given the family history of hypertension and cardiovascular diseases, urgent intervention is re
quired to mitigate potential complications. Diagnostic tests such as ECG and stress test, along with lifestyle modifications emphasizing smoking cessation and dietary
changes, are imperative for 2.4's well-being.
```

- **Mapping Document Example:**

```
Mapping Document
1.1: Ms. Sarah Evans
1.2: 55-year-old
1.3: 321 Elm Street, Chicago, IL
1.4: Sarah
2.1: Mr. William Turner
2.2: 65-year-old
2.3: 654 Birch Street, Boston, MA
2.4: William
```

```
17.1: Mr. Joe DrinkingWater
17.2: 55-year-old
17.3: 55 Penny Lane, Liverpool, UK
17.4: Mr. Joe
18.1: Mr. Alen Maxwell
18.2: aged 60
18.3: 321 Oak Avenue, New York, NY
18.4: Mr. Alen
```

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